

Chapter 7: Student Version

Six questions using a five-word sequence. Use natural logarithms and a calculator. Keep six decimal places while calculating; report final answers to four decimal places.

Data and settings

$$\begin{aligned}\mathcal{V} &= \{\text{three, clever, tired, students, robots, solve, puzzles, [MASK]}\}, \\ z &= (\text{three, clever, students, solve, puzzles}), \quad d = 5, \\ t &= 0.2, \quad \kappa_t = t, \quad h = 0.2, \quad \eta = 0.05.\end{aligned}$$

z is the training target. x is the input constructed from it. The learning rate η updates model parameters; the time step h advances the simulated sequence.

A linear model for every word at every position

$$\ell_j(v) = a_j w_{j,v}, \quad (a_1, a_2, a_3, a_4, a_5) = (1, 1, 3, 1, 1).$$

The supplied features a_j stay fixed. The parameter $w_{j,v}$ is the weight for word v at position j . All entries in this table are trainable weights:

j	three	clever	tired	students	robots	solve	puzzles	[MASK]
1	$\ln 10$	$\ln 6$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$
2	$\ln 4$	$\ln 10$	$\ln 6$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$
3	$\frac{1}{3} \ln 4$	$\frac{1}{3} \ln 4$	$\frac{1}{3} \ln 4$	$\frac{1}{3} \ln 10$	$\frac{1}{3} \ln 6$	$\frac{1}{3} \ln 4$	$\frac{1}{3} \ln 4$	$\frac{1}{3} \ln 4$
4	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 10$	$\ln 6$	$\ln 4$
5	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 4$	$\ln 6$	$\ln 10$	$\ln 4$

For example, $w_{3,\text{students}} = \frac{1}{3} \ln 10$, so $\ell_3(\text{students}) = 3(\frac{1}{3} \ln 10) = \ln 10$.

Predictions use ordinary softmax over all eight words

$$q_{\theta,j}(v \mid x, t) = \frac{e^{\ell_j(v)}}{\sum_{u \in \mathcal{V}} e^{\ell_j(u)}} = p_{1|t}^{\theta}(v \mid x)_j.$$

Here θ contains all the weights $w_{j,v}$. Every vocabulary entry has positive probability at every position. The same notation as the slides is used: q for classifier probabilities, Q for rates, and π for next-step probabilities.

1. Construct the input

The sampled keep indicators are $m = (1, 1, 0, 1, 0)$. Set $x_j = z_j$ when $m_j = 1$, and $x_j = [\text{MASK}]$ otherwise. Write x and the shape of the classifier's full probability output.

This is a numerical teaching model for one input and time. The features and starting weights are supplied so the calculations can be done by hand.

Training: softmax and loss

2. Compute probabilities for all eight words

Calculate $\ell_j(v) = a_j w_{j,v}$ and $e^{\ell_j(v)}$. At position 3, show the calculation for each of the eight words. Then calculate each row's denominator

$$D_j = \sum_{v \in \mathcal{V}} e^{\ell_j(v)}$$

and fill in the complete probability table. Check that each row sums to 1.

j	three	clever	tired	students	robots	solve	puzzles	[MASK]
1								
2								
3								
4								
5								

Working for position 3:

3. Calculate the training loss

Use the token cross-entropy in slide 13 and Algorithm 8:

$$\mathcal{L}(\theta) = - \sum_{j=1}^5 \log q_{\theta,j}(z_j \mid x, t).$$

Write and evaluate all five terms. Does this example have a separate loss term $-\log q_{\theta,3}(\text{robots} \mid x, t)$? Explain using the target z_3 .

Training: gradients and a parameter update

4. Calculate the weight gradients

Use the logit-gradient rule and the chain rule:

$$\frac{\partial \mathcal{L}}{\partial \ell_j(v)} = q_{\theta,j}(v \mid x, t) - \delta_{z_j}(v),$$
$$\frac{\partial \mathcal{L}}{\partial w_{j,v}} = a_j(q_{\theta,j}(v \mid x, t) - \delta_{z_j}(v)).$$

Here $\delta_{z_j}(v)$ equals 1 when $v = z_j$, and 0 otherwise. Calculate all weight gradients. You may group identical values. Show the calculations for students, robots, and [MASK] at position 3 explicitly.

5. Update the weights and recompute predictions

Apply one gradient-descent update to every weight, using the gradients from question 4:

$$w_{j,v}^+ = w_{j,v} - 0.05 \frac{\partial \mathcal{L}}{\partial w_{j,v}}.$$

Use the updated weights to calculate

$$\ell_j^+(v) = a_j w_{j,v}^+, \quad q_{\theta^+,j}(v \mid x, t) = \frac{e^{\ell_j^+(v)}}{\sum_{u \in \mathcal{V}} e^{\ell_j^+(u)}}.$$

Show all updated weights and probabilities in tables on a separate sheet. Repeated values can be calculated once and copied. Calculate $\mathcal{L}(\theta^+)$ and compare it with $\mathcal{L}(\theta)$.

Switch to inference after this update. Keep θ^+ fixed and reuse the same x, t and supplied features for question 6. The rate and sampling formulas use the model predictions and the current sequence x .

Inference: one step with the updated model

6. Convert predictions to rates, then sample

(a) **Schedule.** With $\kappa_t = t$, $t = 0.2$, and $h = 0.2$, calculate

$$\dot{\kappa}_t, \quad c_t = \frac{\dot{\kappa}_t}{1 - \kappa_t}, \quad \alpha_t = hc_t.$$

(b) **Rates.** Use your updated classifier probabilities:

$$Q_t^{\theta^+}(v, j \mid x) = c_t(q_{\theta^+, j}(v \mid x, t) - \delta_{x_j}(v)).$$

Calculate the rate for every word at every position. Show students, robots, and [MASK] at position 3 explicitly. Check that each position's rates sum to zero.

(c) **Next-step probabilities.** Calculate

$$\begin{aligned} \pi_j(v) &= \delta_{x_j}(v) + hQ_t^{\theta^+}(v, j \mid x) \\ &= (1 - \alpha_t)\delta_{x_j}(v) + \alpha_t q_{\theta^+, j}(v \mid x, t). \end{aligned}$$

Check that every entry is nonnegative and each row sums to 1.

(d) **Sample every position.** Use this vocabulary order for cumulative sampling:

(three, clever, tired, students, robots, solve, puzzles, [MASK]).

The independent uniform draws are

$$(u_1, u_2, u_3, u_4, u_5) = (0.55, 0.45, 0.17, 0.60, 0.20).$$

Select each next token, then write X_{t+h} and the new time. Explain why two positions can change in one Algorithm 7 step even though one exact CTMC jump changes one position.

Use a separate sheet for the full rate and probability tables. Source: supplied Chapter 7 notes, pp. 58 and 63–65; teaching slides 12–17.